RESEARCH STATEMENT

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1. Summary of my results

My research concerns the structure of Lagrangian fibration on symplectic varieties.

Definition 1.1. A normal variety X is **symplectic** if it admits a non-degenerate symplectic form $\sigma \in H^0(X^{\mathrm{sm}}, \Omega^2_{X^{\mathrm{sm}}})$ which extends holomorphically onto a resolution of singularities $\pi : Y \to X$. In this setting, a **Lagrangian fibration** is a proper surjective morphism $f : X \to B$ with connected half-dimensional fibers, such that $\sigma|_{X_b} = 0$ for any fiber X_b.

We will need to assume that X is compact for most of our results. Additionally, we will want X to be "irreducible" in a sense, which will be made more precise in (*** link below). To that end, the most general class of symplectic varieties we will work with is the primitive symplectic varieties.

Definition 1.2. A compact symplectic variety X is **primitive symplectic** if it satisfies $H^1(X, \mathcal{O}_X) = 0$ and σ is unique up to scaling, i.e. $H^0(X, \Omega^2_X) = \mathbb{C}\sigma$.

As we will see (*** link below), when X is a primitive symplectic variety admitting a Lagrangian fibration $f : X \to B$, the smooth fibers X_b are Abelian varieties. My main results further classify the structure of the fibers.

Theorem 1.3. *Let X be a primitive symplectic variety of dimension $2n$ admitting a Lagrangian fibration $f : X \to B$. If f is not isotrivial, the Mumford–Tate group of the very general fiber X_b is isogenous to $(\mathrm{Sp}_{2h})^{n/h}$.*

Recall that an Abelian variety is Hodge-generic if it admits no extra Hodge tensors. In this case, it is simple and carries no extra symmetries.

Corollary 1.4. *The very general fiber X_b is isogenous to a product of Abelian varieties $A_1 \times \cdots \times A_{n/h}$, where each A_i is Hodge-generic and A_i is not isogenous to A_j for $i \neq j$.*

The most well-studied class of primitive symplectic varieties consists of hyper-Kähler manifolds, which should be thought of as the natural generalization of K3 surfaces.

Definition 1.5. A compact complex manifold X is **hyper-Kähler** if X is simply connected and its algebra of global holomorphic forms is generated by a non-degenerate symplectic form: $H^0(X, \Omega^{2p}_X) = \mathbb{C}\sigma^p, \quad H^0(X, \Omega^{2p+1}_X) = 0$.

Theorem 1.6. *Let $\mathcal{X}$ be a deformation class of hyper-Kähler manifolds such that*

1. *all members $X \in \mathcal{X}$ have second Betti number $b_2(X) \geq 7$,*
2. *there exists some $X \in \mathcal{X}$ admitting a Lagrangian fibration.*

Choose X to be general among the hyper-Kähler manifolds admitting a Lagrangian fibration, $f : X \to B$. If X_b is a very general fiber of f, then it has Mumford–Tate group $\mathbf{MT}(X_b) = \mathrm{GSp}_{2n}$, i.e. X_b is Hodge-generic.

These results largely came about through understanding the monodromy of the family f. To be more precise, let $f : X^\circ \to B^\circ$ be the restriction of f to its regular points, and let $V = R^1 f_* \mathbb{Z}_{X^\circ}$, viewed as a local system.

Theorem 1.7. *Suppose that X is primitive symplectic, $f : X \to B$ a Lagrangian fibration, $V_\mathbb{C} = V \otimes_\mathbb{Z} \mathbb{C}$ the underlying $\mathbb{C}$-representation of $\pi_1(B^\circ)$.*

(1) *If f is not isotrivial, then $V_\mathbb{C}$ is irreducible.*
(2) *If f is isotrivial, then its smooth fibers are isogenous to E^n, for E an elliptic curve. Furthermore, if K is the endomorphism field of E (possibly $\mathbb{Q}$), then $V_K = U_1 \oplus U_2$ splits into two geometrically irreducible components.*

Remark 1.8. There are a few clarifications to be made regarding the results and contributions. For the non-isotrivial case (1), the later result of Theorem 1.3 determines that a finite-index subgroup of the image of $\pi_1(B^\circ)$ lies in $(\mathrm{Sp}_{2h,\mathbb{Q}})^{n/h}$, and acts on $V_\mathbb{C}$ via a direct sum of standard representations. Such a representation is would not be irreducible, if not for the unaccounted finite part of the monodromy transitively permuting the irreducible factors. For the isotrivial case (ii), the result $X_b \sim E^n$ was first proven in (*** KLM), though we produced an alternative proof of this fact.

2. Background and motivation

Hyper-Kähler manifolds arise naturally in the study of K-trivial varieties. The following result is a classical theorem of Beauville–Bogomolov:

Theorem 2.1 (Beauville–Bogomolov Decomposition Theorem)**.** *Let Z be a compact complex manifold with $\omega_Z \simeq \mathcal{O}_Z$. Then there is a finite étale cover $\pi : Z' \to Z$ such that Z' has the form*

$$Z' = T \times \prod_i X_i \times \prod_j Y_j,$$

where T is a complex torus, the X_i are hyper-Kähler manifolds, and the Y_j are irreducible Calabi–Yau manifolds.

This theorem has since been generalized to allow for Z to have klt singularities (Bakker–Guenancia–Lehn); in this case, the X_i are irreducible holomorphic symplectic varieties, a condition stronger than primitive symplectic but weaker than hyper-Kähler.

Thus the problem of understanding families of K-trivial varieties boils down to understanding families of complex tori, hyper-Kählers, and ICYs. The former is largely classical, while the latter is also somewhat well-studied, thanks to work of Laza–Friedman (loc. cit.***) In contrast, we have a relatively poor understanding of the deformation classes of hyper-Kähler manifolds.

Given a polarized hyper-Kähler manifold (X, H), one can construct its deformation space as a type IV Shimura variety: take $L = H^\perp \subsetneq H^2(X, \mathbb{Z})$, define

$$\mathcal{D}_H = \{[\sigma] \in \mathbb{P}L_\mathbb{C} \mid \sigma \wedge \sigma = 0, \quad \sigma \wedge \overline{\sigma} > 0\}.$$

Then the deformation space of X is constructed as

$$\mathcal{X} = \Gamma \backslash \mathcal{D} = \Gamma \Big\backslash \tfrac{SO(2,b_2-3)}{SO(2)\times SO(b_2-3)},$$

where $b_2 = \dim H^2(X, \mathbb{C})$.

As such, to constuct a deformation space of hyper-Kähler manifold, it suffices to exhibit one member. There are currently four known deformation types: the infinite families $K3^{[n]}$ and Kum_n, occuring in every even dimension $2n$, and the two sporadic types $OG6$ and $OG10$ (loc. cit. O'Grady), occuring in dimensions 6 and 10 respectively.

Now consider the following central conjecture.

Conjecture 2.2. *(Hyper-Kähler SYZ Conjecture) Let $\mathcal{X}$ be a deformation space of hyper-Kähler manifolds. There exists some $X \in \mathcal{X}$ which admits a Lagrangian fibration $f : X \to B$.*

If true, one could make statements about all deformation classes of hyper-Kählers by simply proving them for those admitting Lagrangian fibrations. A recent paper of (engel 25***) assumed the SYZ conjecture to show that there are only finitely many deformation classes of hyper-Kählers in any fixed dimension.

As such, we believe it important to understand the structure of Lagrangian fibrations. Incidentally, it turns out that they carry many constraints.

Theorem 2.3. *Let X be a hyper-Kähler manifold of dimension $2n$, $f : X \to B$ a proper surjective morphism with connected fibers, $B \neq \{p\}$. The following all hold.*

[Greb Lehn 14] f is a Lagrangian fibration.
[Matsushita 99] The smooth fibers of f are Abelian varieties.
[Matsushita 14] B is $\mathbb{Q}$-Fano, $\mathbb{Q}$-factorial, and klt with Picard rank 1. There is an isomorphism $\mathrm{IH}^*(B, \mathbb{Q}) \simeq H^*(\mathbb{P}^n, \mathbb{Q})$.
[Hwang 08] If B is smooth, then $B \simeq \mathbb{P}^n$.
[Bakker] If f is not isotrivial, then it varies maximally, i.e. the period map $\mathcal{P} : B^\circ \to \mathcal{A}_{n,\eta}$ has an n-dimensional image.

My results Corollary 1.4, Theorem 1.6 aim to strenthen (2). Furthermore, as we will see in §3, this may yet have consequences for (3). The base B is a common area of study in this field, due to the following conjecture of Matsushita, partially resolved due to Hwang in (4).

Conjecture 2.4 (Matsushita)**.** *Let X be a hyper-Kähler manifold, $f : X \to B$ a Lagrangian fibration. Then $B \simeq \mathbb{P}^n$.*

3. Future directions

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